

Module 4

Telecommunication Devices and Standards

Syllabus

- *Telephone, Mobile Phones, Modems*
 - *Telecommunication standards (ITU, IEEE, 3GPP)*
 - *Basics of 2G, 3G, 4G, and 5G*
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4.1 Key Telecommunication Devices

Overview

This unit examines the fundamental hardware interfaces that act as endpoints in a telecommunication network. These devices form the endpoints of communication systems, enabling voice, data, and multimedia exchange. Their development closely follows the evolution of telecommunication networks—from analog, circuit-switched systems to digital, packet-switched, high-speed networks. The study focuses on the evolution, working principles, and technical architectures of three primary devices: the Telephone, the Mobile Phone, and the Modem.

4.1.1 Telephone

A telephone set (or subscriber station) is the apparatus used by a subscriber to interface with the Public Switched Telephone Network (PSTN). To provide effective two-way voice communication, a standard telephone set must perform several essential functions such as signal conversion, signalling, addressing, alerting, powering etc.

Block Diagram of a Telephone Set

The functional block diagram of a standard telephone set is shown in the figure.

- **Line Interface (Tip and Ring):** The telephone set is connected to the local exchange through a two-wire line commonly referred to as the Tip and Ring conductors. These two wires carry both DC supply and voice signals. The exchange provides a nominal DC voltage of approximately –48 V across the line when the telephone is in the on-hook condition. When the

handset is lifted, a closed loop is formed and current begins to flow, which signals the exchange that the subscriber is requesting service. The same pair of wires is used for transmitting and receiving voice signals, making it a two-wire communication system.

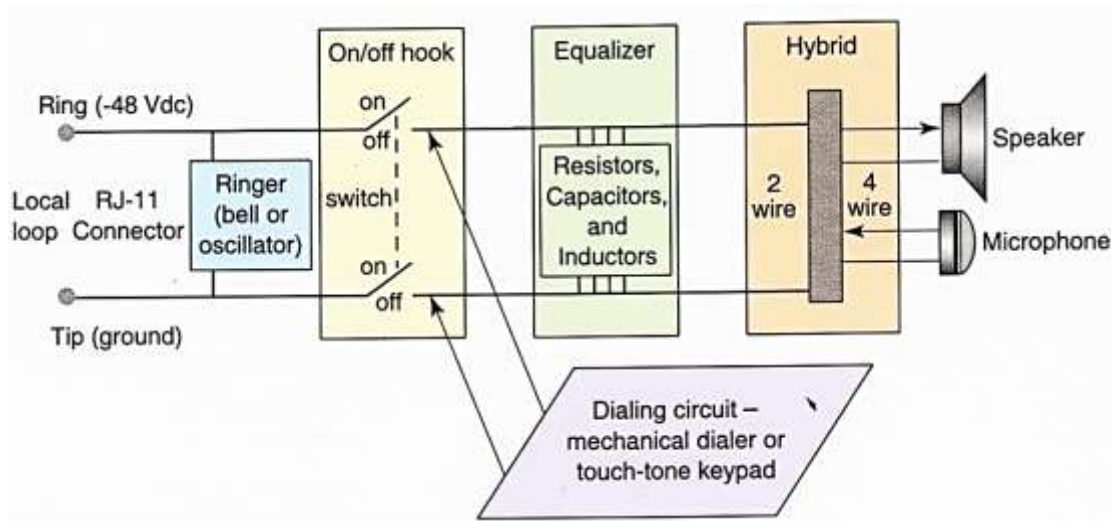


Fig. Functional block diagram of a standard telephone set

- **Ringer Circuit:** The ringer circuit is responsible for producing an audible ringing sound when an incoming call is received. In traditional telephone sets, an electromechanical bell is used, while modern sets use electronic ringers. When the exchange sends ringing signal, the ringer circuit activates and alerts the subscriber of an incoming call.
- **Hook Switch:** The hook switch is a mechanical or electronic switch that operates when the handset is lifted or placed back on the cradle. In the on-hook condition, the switch remains open and no loop current flows through the telephone circuit. When the handset is lifted (off-hook condition), the switch closes, completing the circuit between the telephone set and the exchange. This allows loop current to flow, which informs the exchange that the subscriber wants to initiate or answer a call. Thus, the hook switch performs the important function of connecting and disconnecting the telephone from the network.
- **Dialing Circuit:** The dialing circuit is used to transmit the called party's number to the exchange. In older rotary dial telephones, dialing is achieved using pulse dialing, where the dial mechanism generates a series of make-and-break pulses corresponding to each digit. In modern push-button telephones, Dual Tone Multi-Frequency (DTMF) signaling is used.

- **Microphone (Transmitter):** The microphone, also called the transmitter, acts as an input transducer. It converts human speech into corresponding electrical signals. In traditional telephone sets, a carbon microphone is used. It contains carbon granules whose resistance changes when sound pressure is applied. As the user speaks, variations in sound pressure cause changes in resistance, which in turn vary the current flowing through the circuit. This produces an electrical signal that is an analog representation of the speech waveform.
- **Receiver (Earpiece):** The receiver, or earpiece, performs the reverse function of the microphone. It converts incoming electrical signals into audible sound waves. Typically, it operates on the electromagnetic principle. When a varying electrical current flows through a coil inside the receiver, it creates a varying magnetic field. This magnetic field moves a diaphragm back and forth in accordance with the electrical signal, thereby producing sound waves that replicate the original speech signal transmitted by the calling party.
- **The Equalizer Circuit:** The equalizer circuit is a combination of passive components like resistors, capacitors, and inductors used to regulate the strength and quality of voice signals. Its primary job is to ensure that the volume and clarity of the call remain consistent, regardless of the length or resistance of the telephone cable. Essentially, it balances the transmitter and receiver efficiency so that the sound doesn't become too loud on short lines or too weak on long ones.
- **Hybrid Network (Hybrid Transformer):** The hybrid network is an essential component of a telephone set that enables two-way communication over a two-wire line. Since the same pair of wires is used for both transmission and reception, the hybrid circuit separates the transmitted signal from the received signal. It effectively converts the two-wire circuit into an equivalent four-wire circuit internally, providing separate paths for transmit and receive signals. One of its important functions is to reduce sidetone, which is the portion of the speaker's own voice heard in the receiver. Proper balancing of the hybrid network improves voice clarity and prevents excessive echo.

Functions of the Telephone Set

A telephone set performs several important functions to enable reliable two-way voice communication over the Public Switched Telephone Network (PSTN). These functions ensure proper signaling, voice transmission, call control, and user interaction with the telephone network.

- **Notification of Incoming Call:** The telephone set notifies the subscriber of an incoming call through an audible signal such as a bell or electronic ringer, or sometimes a visible indicator such as a flashing light. This alerting signal is intentionally noticeable so that the user responds promptly. The ringing signal is generated by the central office and activates the ringer circuit in the telephone.
- **Indication of Call Answer (Off-Hook Signaling):** When the subscriber lifts the handset, the telephone provides a signal to the central office indicating that the call has been answered. This is achieved by closing the loop circuit (off-hook condition), allowing loop current to flow. The central office detects this current and connects the call.
- **Conversion of Speech Energy:** The telephone set converts acoustical (sound) energy into electrical energy at the transmitter (microphone) and converts electrical energy back into sound at the receiver (earpiece). The microphone first converts sound waves into mechanical vibrations and then into corresponding electrical signals. The receiver performs the reverse operation, enabling two-way voice communication.
- **Sending Dialing Information:** The telephone set provides a method for entering and transmitting destination telephone numbers to the central office over the local loop. This is accomplished using rotary dialing (pulse dialing) or push-button dialing (DTMF – Dual Tone Multi-Frequency signaling). The dialing information allows the switching system to route the call correctly.
- **Regulation of Speech Signal Amplitude:** The telephone set regulates the amplitude of the transmitted speech signal to maintain consistent voice levels. This prevents excessive signal strength that could cause distortion or interfere with other conversations in adjacent cable pairs (crosstalk). Circuits such as equalizers and transmit regulation networks help maintain proper signal levels.
- **Request for Outgoing Call (Dial Tone Initiation):** When a subscriber wishes to place an outgoing call, lifting the handset (off-hook) informs the central office that service is required. The switching system then responds by sending a dial tone, indicating that the subscriber may begin dialing the desired number.
- **Sidetone Control:** The telephone set allows a small portion of the transmitted signal to be fed back into the receiver. This feedback signal, called sidetone or talkback, enables the speaker to hear their own voice at a low level. Sidetone prevents users from speaking too loudly and improves the natural feel of conversation.

- **On-Hook and Off-Hook Control:** When the telephone is not in use (on-hook), it provides an open circuit to the local loop, indicating an idle condition. When the handset is lifted (off-hook), the circuit closes, creating a busy condition and allowing current to flow. This mechanism informs the central office about the status of the subscriber line.
- **Transmission and Reception of Call Progress Signals:** The telephone set enables the transmission and reception of various signaling tones and call progress indicators between the subscriber and the central office. These include dial tone, ringing tone, busy tone, dial pulses, DTMF signals, and on-hook/off-hook signaling. These signals ensure proper call setup, supervision, and termination.

4.1.2 The Mobile Phone (Cellular Phone)

A mobile phone is a portable, wireless telecommunication device that enables voice, messaging, and data communication over a cellular network, allowing users to communicate while moving across wide geographic areas.

Mobile phones use radio frequency (RF) communication instead of physical wires and operate based on the cellular concept.

Internal Block Diagram of a Basic Mobile Phone (Mobile Station – MS)

The internal structure of a basic mobile phone handset can be divided into two main sections: (1) Transmitter Section and (2) Receiver Section. In addition to above the two sections, a mobile phone also contains some supporting and system control blocks.

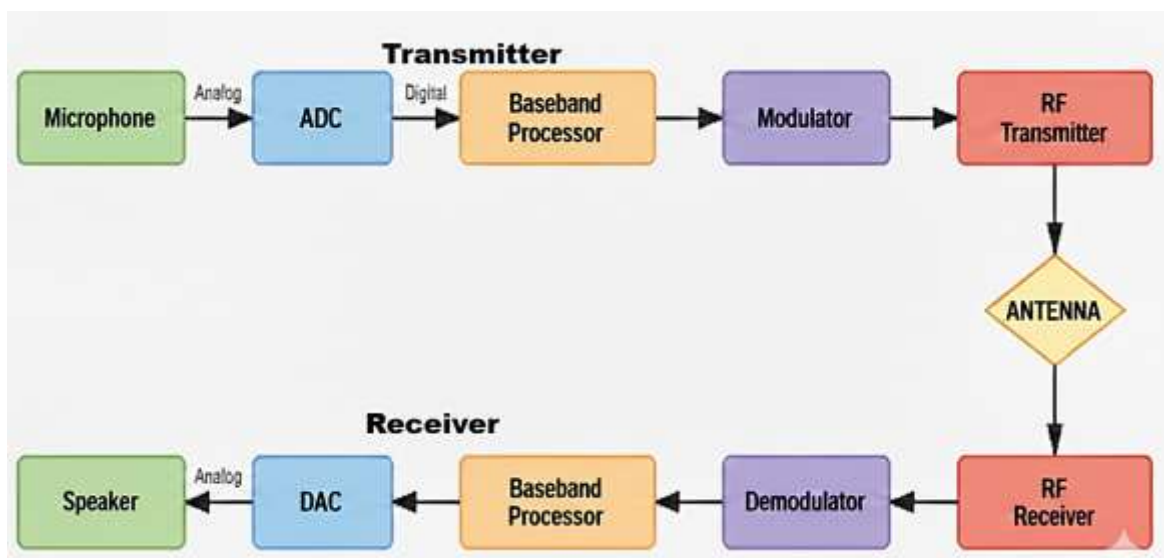


Fig. Block diagram of a basic mobile phone

Transmitter Section (Uplink)

- **Microphone:** The microphone acts as an input transducer. It converts the user's acoustic sound waves (analog voice signals) into corresponding electrical analog signals. This is the first stage of mobile communication.
- **Analog-to-Digital Converter (ADC):** The ADC converts the analog electrical voice signal into a digital bitstream. This conversion is essential because modern cellular communication systems process and transmit data in digital form.
- **Baseband Processor (Transmitter Side):** The baseband processor is the main communication control unit of the mobile phone. It processes the digital voice or data before transmission. It compresses the signal, adds error-correction bits for reliable communication, and encrypts it for security. Finally, it formats the data according to the cellular standard before sending it to the modulator.
- **Modulator:** The modulator combines the processed digital data with a high-frequency carrier signal to produce a radio frequency (RF) signal. The modulation technique depends on the technology, such as GMSK in GSM, OFDM in 4G, and advanced OFDM variants in 5G systems.
- **RF Transmitter (Power Amplifier):** The RF transmitter section mainly consists of a Power Amplifier (PA). It increases the strength of the modulated RF signal to a suitable level so that it can travel the required distance. The transmitter operates in assigned frequency bands such as 900 MHz or 1800 MHz.
- **Antenna (Transmit Mode):** The antenna converts the amplified electrical RF signal into electromagnetic waves and radiates them into free space toward the nearest Base Transceiver Station (BTS).

Receiver Section (Downlink)

- **Antenna (Receive Mode):** In receive mode, the antenna captures electromagnetic waves transmitted by the BTS and converts them into weak electrical RF signals.
- **RF Receiver (Low-Noise Amplifier):** The RF receiver contains a Low Noise Amplifier (LNA). It amplifies the weak incoming RF signal while introducing minimal internal noise. It also filters out unwanted frequencies and interference.
- **Demodulator:** The demodulator removes the high-frequency carrier from the received signal and extracts the original digital baseband information. This process is the reverse of modulation.

- **Baseband Processor (Receiver Side):** The baseband processor on the receiving side processes the incoming digital signal after demodulation. It removes encryption to restore the original data and performs error correction to ensure accurate communication. It then decodes the signal and reconstructs the digital voice samples before sending them to the DAC.
- **Digital-to-Analog Converter (DAC):** The DAC converts the processed digital voice signal back into a continuous analog electrical signal because the speaker operates on analog signals.
- **Speaker:** The speaker acts as an output transducer. It converts the analog electrical signal into sound waves so that the user can hear the caller's voice clearly.

Supporting and System Control Blocks

- **SIM Interface:** Interfaces with the SIM card to read subscriber information, authentication keys, and phone number for network identification.
- **Display & Keypad/Touchscreen:** Provides user interface for input (touchscreen/keypad) and output (display screen) to interact with the phone.
- **Power Management Unit (Battery):** Supplies and regulates power to all components. Manages battery charging, voltage regulation, and power distribution.
- **Memory (RAM/ROM):** RAM stores temporary data and running applications. ROM/Flash memory stores the operating system, applications, and user data permanently.

Working Principle of a Mobile Station (MS)

A Mobile Station (MS), commonly called a mobile phone or handset, is a wireless communication device used in cellular networks. It enables two-way communication by transmitting and receiving radio signals from the nearest Base Transceiver Station (BTS).

The mobile station operates on the principle of a full-duplex radio transceiver, meaning it can transmit and receive signals simultaneously.

The entire working process can be divided into three main parts: (1) Transmission Process (Uplink), (2) Reception Process (Downlink) and (3) System Coordination and Control.

Transmission Process (Uplink Path)

The uplink path describes how a user's voice is transmitted from the mobile phone to the cellular network.

- **Signal Capturing and Digitization:** When a user speaks into the microphone, sound waves are converted into an analog electrical signal. This analog signal is then converted into digital form by the Analog-to-Digital Converter (ADC). The output is a stream of binary data representing the voice signal.
- **Baseband Processing:** The digital signal is processed by the baseband processor. First, speech coding compresses the signal to reduce the required bandwidth. Then, channel coding adds redundant bits for error detection and correction to protect the signal from noise and interference. The signal is also encrypted to ensure secure communication.
- **Modulation and Up-Conversion:** After processing, the digital data is passed to the modulator. Here, the digital bits are superimposed onto a high-frequency carrier wave. Depending on the cellular technology, appropriate modulation techniques are used. This step converts baseband data into a radio frequency (RF) signal suitable for wireless transmission.
- **RF Amplification and Radiation:** The modulated RF signal is amplified by the RF transmitter to the required power level. Finally, the antenna converts the electrical RF signal into electromagnetic waves and radiates them toward the nearest BTS.

Reception Process (Downlink Path)

The downlink path describes how the mobile station receives and processes signals from the BTS.

- **Signal Capture:** The antenna of the mobile phone receives weak electromagnetic waves transmitted by the BTS. These waves are converted into electrical RF signals.
- **Low Noise Amplification:** Since the received signal is very weak, it is amplified by a Low Noise Amplifier (LNA). The LNA increases signal strength while minimizing additional noise.
- **Demodulation:** The amplified RF signal is passed to the demodulator, which extracts the original digital data from the carrier wave. This process separates the information signal from the high-frequency carrier.
- **Baseband Decoding:** The baseband processor then performs channel decoding to detect and correct transmission errors. It also decrypts the

signal to restore the original data. Finally, speech decoding reconstructs the digital voice signal.

- **Digital-to-Analog Conversion and Output:** The processed digital voice signal is converted back into an analog electrical signal using a Digital-to-Analog Converter (DAC). This analog signal drives the speaker, which converts it into sound waves that can be heard by the user.

System Coordination and Control Functions

In addition to transmission and reception, the mobile station performs several supporting functions.

- **Network Authentication:** The Subscriber Identity Module (SIM) stores subscriber identity information and security keys. It continuously provides authentication data to the processor to ensure secure and authorized access to the network.
- **Power Management:** The Power Management Unit (PMU) regulates battery usage and controls transmission power. The RF transmission power is automatically adjusted based on signal strength and distance from the BTS. This improves battery life and reduces interference.
- **Full-Duplex Operation:** Separate frequency bands or time slots are used for uplink and downlink communication. This allows the mobile station to transmit and receive simultaneously without interference.

4.1.3 Modem

Modem is a contraction of the terms MOdulator and DEmodulator. It is a device that converts digital data (0s and 1s) from a computer into analog signals (varying waveforms) for transmission over a channel, and vice versa. A modem enables a computer or other digital device to communicate over analog transmission media, such as traditional telephone lines or cable systems. It acts as a bridge between the digital world of computers and the analog world of telecommunication channels.

Functions of a Modem

- **Modulation:** The process of converting digital bits from a computer into an analog signal by varying the amplitude, frequency, or phase of a carrier wave.
- **Demodulation:** The process of receiving an incoming analog signal and converting it back into the original digital data format for the computer to process.

- **Signal Conditioning:** Modems perform filtering and amplification to ensure the signal can travel long distances without significant degradation.
- **Error Detection:** Modern modems include protocols to check if the data was corrupted during transmission and request retransmission if necessary.
- **Data Compression:** They often compress data before transmission to increase the effective "throughput" or speed of the connection.

Block Diagram of a Modem

A Modem consists of two primary sections: the Transmitter Section which prepares data for the channel, and the Receiver Section which retrieves the original data.

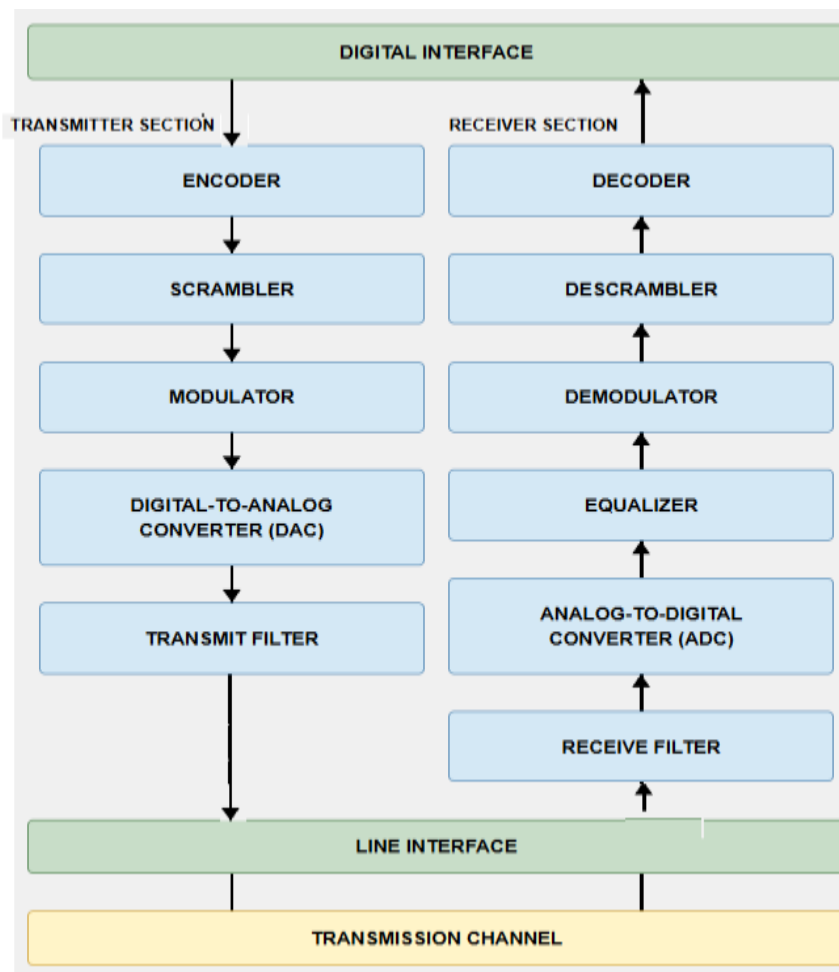


Fig. Block Diagram of a Modem

Common Interfaces

- **Digital Interface:** This is the entry and exit point for data from the computer or terminal. It manages the flow of digital bits (0s and 1s) between the DTE (Data Terminal Equipment) and the modem.
- **Line Interface:** This block acts as the physical connection to the transmission medium.
- **Transmission Channel:** The physical medium, such as a telephone line (PSTN), coaxial cable, or wireless link, that carries the analog signal.

Transmitter Section (Digital to Analog)

This section is responsible for converting the computer's digital data into a format suitable for long-distance analog transmission.

- **Encoder:** It converts the raw data bits into a specific format or code. This stage often includes error-correction coding to help the receiver detect or fix bits corrupted during transmission.
- **Scrambler:** It re-arranges the data bits into a seemingly random pattern. This is done to prevent long sequences of 0s or 1s, which helps in maintaining synchronization and ensures a constant power spectrum across the signal.
- **Modulator:** The core component that performs the mathematical process of modulation. It maps input digital bits onto a digital representation of a carrier wave by calculating the required changes in amplitude, frequency, or phase. It produces a high-speed stream of digital samples that describe the analog wave to be generated.
- **Digital-to-Analog Converter (DAC):** It converts the discrete modulated digital values into a continuous, smooth analog voltage or current waveform.
- **Transmit Filter:** It limits the signal's bandwidth to ensure it stays within the allocated frequency range of the channel. This prevents interference with other communication services (adjacent channel interference).

Receiver Section (Analog to Digital)

This section reverses the transmitter's processes to extract the original data from the noisy received signal.

- **Receive Filter:** It filters out noise and signals from other channels that may have been picked up during transmission, allowing only the desired signal to pass.

- **Analog-to-Digital Converter (ADC):** It samples the incoming analog waveform and converts it back into a digital numerical format for processing.
- **Equalizer:** This block compensates for distortions introduced by the transmission channel.
- **Demodulator:** It "reads" the modulated carrier wave to determine which digital state (0 or 1) was originally sent, effectively stripping the data from the carrier.
- **Descrambler:** It reverses the scrambling process performed at the transmitter, restoring the data to its original sequence.
- **Decoder:** It translates the coded bits back into the original message format and performs error checking to ensure the data is accurate.

Working Principle

The working principle of a modem is based on Signal Conversion:

- **Digital-to-Analog Transition:** Computers communicate using discrete electrical pulses (0V for '0' and 5V for '1'). However, these pulses cannot travel through analog channels like phone lines because of frequency limitations. The modem's Modulator translates these pulses into a continuous wave (Analog) that is compatible with the channel's characteristics.
- **Transmission:** The analog signal travels through the wired (copper/fiber) or wireless medium.
- **Analog-to-Digital Transition:** At the receiving end, the signal is often weak and noisy. The modem's Demodulator "reads" the characteristics of the incoming wave (like its frequency or phase) to determine whether the sender sent a '0' or a '1', thus recreating the digital file.

Applications of Modems

- **Internet Connectivity:** Connecting home or office networks to an Internet Service Provider (ISP) via DSL, Cable, or Fiber-optic lines.
- **Remote Access:** Allowing employees to connect to a central company server from a remote location.
- **Point-of-Sale (POS) Terminals:** Used in retail stores to transmit credit card authorization data over telephone or cellular networks.

- **Industrial Automation:** Used in SCADA systems to monitor and control remote industrial equipment like pipelines or power grids.
- **Facsimile (Fax):** Traditional fax machines use built-in modems to send document images over PSTN lines.

Types of Modems

- **Dial-up Modem:** Uses PSTN, low speed (historical)
- **DSL Modem:** (DSL – Digital Subscriber Line) Uses standard telephone lines but operates at higher frequencies (above voice band), allowing always-on internet alongside voice. Uses a splitter, Broadband over telephone lines.
- **Cable Modem:** Uses cable TV (coaxial) network.
- **Wireless Modem:** Often called a "Dongle" or "Hotspot," it connects to cellular (4G/5G) networks to provide internet access to other devices via Wi-Fi or USB.
- **Fiber Modem (ONT):** (ONT – Optical Network Terminal) Converts optical signals for fiber internet. The "modem" for Fiber-to-the-Home (FTTH). Converts optical signals to electrical signals.

Advantages and Limitations of Modems

- **Advantages:** Enables internet access over diverse media, facilitates long-distance digital communication, available in wired and wireless forms.
- **Limitations:** Speed and quality are dependent on underlying line technology and service provider. Performance affected by noise and line quality.

4.2 Telecommunication Standards

Telecommunication standards are internationally agreed-upon technical specifications that define the rules for communication systems. They ensure interoperability, reliability, and global connectivity by standardizing parameters like frequency usage, signal formats, protocols, and interfaces. Without these standards, devices from different manufacturers or networks in different countries would be unable to communicate, leading to a fragmented and inefficient global system.

Need for Telecommunication Standards

The implementation of global standards is driven by several industrial and economic necessities:

- **Interoperability:** Interoperability is the primary purpose of telecommunication standards. It ensures that devices and systems from different manufacturers can communicate seamlessly with each other. For example, a smartphone manufactured from various vendors (e.g., Nokia, Samsung) to function across different carriers (e.g., AT&T, Jio).
- **Global Compatibility:** Enables international roaming and seamless data exchange across borders. Standards enable global roaming and international communication services. Without standardization, mobile phones would not work across different countries, and international calls would be impossible. Standards ensure that networks worldwide use compatible technologies and protocols.
- **Mass Production & Cost Reduction:** Standardization allows for economies of scale, reducing the cost of components for manufacturers and consumers.
- **Technological Evolution:** Provides a stable technical foundation upon which companies can build new features (like 6G) without disrupting existing legacy systems.
- **Safety & Quality:** Ensures reliable and safe operation of communication systems.
- **Quality of Service** – Standards define minimum performance requirements, ensuring that telecommunication services meet acceptable quality levels. This includes parameters like data rates, latency, reliability, and coverage.

Without standards, communication systems would be incompatible, fragmented, and unreliable.

Major Telecommunication Standardization Bodies

The three important organizations in telecommunication standardization are:

- International Telecommunication Union (ITU)
- Institute of Electrical and Electronics Engineers (IEEE)
- 3rd Generation Partnership Project (3GPP)

4.2.1 International Telecommunication Union (ITU)

The International Telecommunication Union (ITU) is a specialized agency of the United Nations (UN) and is the oldest global telecommunication organization, established in 1865.

Main Functions of ITU

- **Global Coordination:** ITU coordinates the shared global use of radio frequency spectrum, satellite orbits, and telecommunication standards to prevent interference and ensure efficient utilization of resources.
- **Standards Development:** ITU develops international standards (known as Recommendations) for all areas of telecommunication, ensuring global interoperability and quality. For example, ITU developed 'IMT-Advanced' for 4G and 'IMT-2020' for 5G.
- **Spectrum Management:** ITU manages the radio frequency spectrum and satellite orbital positions through international agreements and frequency allocation tables.
- **Development Initiatives:** ITU assists developing countries in building their telecommunication infrastructure and bridging the digital divide through technical assistance and capacity building programs.
- **Policy and Regulatory Guidance:** ITU provides policy frameworks and regulatory guidance to help nations develop their telecommunication sectors effectively.

Sectors of ITU

The ITU operates through three main sectors:

- **ITU-R (Radio-communication):** Manages the international radio-frequency spectrum and satellite orbits. Prevents interference and allocates frequency bands through the World Radiocommunication Conference (WRC). Example: Allocating the 3.5 GHz band for global 5G use.
- **ITU-T (Telecommunication Standardization):** Develops "Recommendations" for wired networks and core systems. Examples include the G-series (voice coding), H-series (video compression like H.264), and V-series (modem standards).
- **ITU-D (Telecommunication Development):** Focuses on bridging the digital divide in developing countries. It promotes telecom development in developing countries, provides technical assistance and policy guidance,

promotes digital inclusion and bridging the digital divide, provides technical assistance and training programs, facilitates knowledge sharing and best practices

Importance of ITU

- Enables international calls and roaming.
- Provides global regulatory and spectrum frameworks.
- Defines global requirements for mobile generations (IMT-2000, IMT-Advanced, IMT-2020).

4.2.2 Institute of Electrical and Electronics Engineers (IEEE)

The IEEE is the world's largest professional technical organization and a non-governmental body. It develops standards through an open, consensus-based process and primarily governs Local Area Networks (LAN) and Metropolitan Area Networks (MAN) and short-range wireless technologies. IEEE standards are widely adopted as de facto global standards.

Role of IEEE in Communication Systems

- **Standards Development:** IEEE develops standards through its Standards Association (IEEE-SA). These standards cover electrical and electronic systems, including communications, computing, and power systems. IEEE defines physical layer standards, MAC layer protocols and data transmission formats. For example, wi-fi devices such as a router and laptop from different companies can communicate successfully because both follow the IEEE 802.11 standard.
- **Innovation and Research:** IEEE publishes research papers, journals, and conference proceedings that advance the state of the art in communication technologies.
- **Professional Development:** IEEE provides educational resources, certifications, and professional networking opportunities for engineers and researchers.
- **Industry Collaboration:** IEEE brings together academia, industry, and government to develop practical, implementable standards.

IEEE 802 Family of Standards

The most important IEEE telecom standards belong to the IEEE 802 series, which primarily define Physical (PHY) and Data Link (MAC) layer technologies.

The 802 series represent the most significant IEEE standards in data communication:

- **IEEE 802.3 (Ethernet):** The standard for wired LANs using twisted pair or fiber optics.
- **IEEE 802.11 (Wi-Fi):** The global standard for Wireless LAN (WLAN). Versions include 802.11ac (Wi-Fi 5) and 802.11ax (Wi-Fi 6).
- **IEEE 802.15 (WPAN):** Standards for Wireless Personal Area Networks, most notably 802.15.1 (Bluetooth).
- **IEEE 802.16 (WiMAX):** Designed for broadband wireless metropolitan access.

Role and Importance of IEEE

- Ensures high-speed data communication.
- Enables interoperability in computer networks.
- Forms the backbone of internet connectivity in homes, offices, and campuses.

4.2.3 3rd Generation Partnership Project (3GPP)

The 3GPP is a collaborative partnership between regional standards organizations (e.g., ETSI in Europe, ATIS in North America). It is the central body responsible for defining the complete technical specifications for cellular mobile communication.

Role in Mobile Communication

- **Technical Specification Development:** 3GPP develops complete technical specifications for: radio access technologies (air interface), core network architecture, service and system aspects, terminals and devices, testing and conformance
- **Evolution of Mobile Technologies:** 3GPP has been responsible for standardizing all major cellular technology generations such as 2G (GSM enhancements), 3G (UMTS), 4G (LTE), 5G (5G NR)
- **Ensuring Global Compatibility:** 3GPP standards enable global roaming (users can use their phones internationally), multi-vendor ecosystems (equipment from different manufacturers works together), economy of scale (single global standard reduces costs), technology transfer (all countries can implement the same proven technology)

Scope and Generations

3GPP manages the complete technical specifications for mobile networks, including the Radio Access Network (RAN) and Core Network.

- **2G:** GSM, GPRS, EDGE.
- **3G:** UMTS, WCDMA, HSPA.
- **4G (LTE):** Long Term Evolution, introducing an all-IP packet core.
- **5G (5G-NR):** New Radio, focusing on ultra-low latency and massive IoT connectivity.

Working Model

- 3GPP releases specifications in sequential "Releases." For example, Release 15 defined the first functional 5G standard, while subsequent releases introduce features like Network Slicing and 6G research.

Importance of 3GPP

- Ensures mobile phones work across countries and operators
- Enables international roaming and seamless mobility
- Every commercial cellular network follows 3GPP specifications

4.2.4 Comparative Summary of ITU, IEEE, and 3GPP

Parameter	ITU	IEEE	3GPP
Full Form	International Telecommunication Union	Institute of Electrical and Electronics Engineers	Third Generation Partnership Project
Nature	UN Agency (Governmental)	Professional Society (NGO)	Industry Consortium
Role	Regulatory & Global Coordination	Develops networking standards	Develops cellular standards
Primary Focus	Global Regulation & Infrastructure (Spectrum, Framework)	Networking & Local Connectivity (LAN, WLAN)	Cellular Mobile Systems (3G/4G/5G)
Examples of Standards	IMT-2000, IMT-2020	802.3, 802.11	UMTS, LTE, 5G NR
Applications	Spectrum, policy, telecom framework	Ethernet, Wi-Fi, Bluetooth	Mobile communication, IoT

How They Work Together – The Smartphone Example:

1. The ITU allocates the spectrum band (e.g., 3.5 GHz) and sets the high-level requirements for what constitutes "5G" (IMT-2020).
2. 3GPP writes the detailed rulebook for how a smartphone's cellular modem and the mobile network must operate within that spectrum to deliver 5G services.
3. The IEEE defines the local wireless standards (Wi-Fi, Bluetooth) the phone uses when not on the cellular network.

Together, these organizations create the layered technical and regulatory ecosystem that ensures seamless, global communication, allowing a device purchased anywhere to work almost everywhere.

4.3 Basics of 2G, 3G, 4G, and 5G Mobile Communication Systems

The evolution of mobile communication represents the transformation of wireless networks from basic digital voice systems to high-speed intelligent broadband networks. Each generation (G) represents a major technological upgrade in terms of data speed, switching techniques, and the types of services supported.

4.3.1 Second Generation (2G)

2G (Second Generation) is the first digital cellular technology, introduced in the early 1990s. It marked a fundamental shift from analog (1G) to digital transmission, enabling more efficient use of spectrum.

Key Features and Advancements:

- **Digital Voice Transmission:** Used digital signals instead of analog, resulting in clearer voice quality and reduced interference.
- **Security:** Introduced digital encryption for call privacy.
- **SMS (Short Message Service):** Enabled the first text messaging capabilities.
- **International Roaming:** Standardization allowed users to use their devices across different national networks.

Technical Standards:

- **GSM (Global System for Mobile Communications):** Most widely used standard, employing TDMA technology.

- **GPRS / EDGE:** Known as 2.5G/2.75G, these introduced basic packet-switched data to the 2G core.

Limitations:

- **Low Data Speeds:** Peak speeds were limited to ~384 kbps (EDGE).
- **Circuit-Switched Focus:** Designed primarily for voice; highly inefficient for modern internet browsing.

4.3.2 Third Generation (3G)

3G represents a leap in mobile communications introduced in the early 2000s, designed to handle high-speed data alongside voice, marking the beginning of the smartphone era.

Key Features and Advancements:

- **Mobile Broadband:** Provided data speeds sufficient for web browsing and emails.
- **Packet-Switched Data:** Implemented more efficient data transmission compared to the circuit-switched nature of 2G.
- **Video Calling:** Enabled the first real-time video telephony capabilities.

Technical Standards:

- **UMTS / WCDMA:** The primary standards for 3G, offering higher spectral efficiency.

Services and Applications:

- Web browsing, GPS navigation, basic mobile apps, and video conferencing.

Limitations:

- **High Latency:** Network delays (100–500 ms) were too high for real-time gaming or HD streaming.
- **Bandwidth Constraints:** Insufficient for the "data explosion" caused by modern social media and HD video.

4.3.3 Fourth Generation (4G)

4G is an All-IP based network that provides ultra-broadband internet access. Introduced in the late 2000s, it completely eliminated circuit-switching, treating voice as just another form of data.

Key Characteristics of 4G LTE:

- **LTE (Long Term Evolution):** The global standard for 4G, redesigning the network for high speed and low latency.
- **OFDMA Technology:** Orthogonal Frequency Division Multiple Access for efficient spectrum use.
- **VoLTE (Voice over LTE):** Technology that allows HD voice calls over the 4G data network instead of traditional voice channels.
- **Latency:** Significantly reduced to 10–20 ms.

Architecture Components:

1. **eNodeB (Base Station):** Handles radio communications with the mobile device.
2. **EPC (Evolved Packet Core):** The all-IP backbone that manages data traffic and mobility.

Services and Applications:

- HD video streaming (Netflix/YouTube), Cloud computing, and high-speed online gaming.

Limitations:

- **Device Density:** Cannot handle the massive number of devices (IoT) required for smart cities.
- **Latency:** While low, it is still too high for "mission-critical" tasks like remote surgery.

4.3.4 Fifth Generation (5G)

5G is the most advanced wireless technology, designed to connect virtually everyone and everything, including machines and IoT devices.

Key Features and Advancements:

- **Ultra-High Speed:** Peak speeds up to 20 Gbps (10–100x faster than 4G).
- **Ultra-Low Latency:** Reduced to 1 millisecond, enabling real-time critical responses.
- **Network Slicing:** Creating virtual "slices" of the network tailored for specific uses (e.g., a slice for emergency services).
- **Massive MIMO:** Large antenna arrays to improve capacity and coverage.

Services and Applications

- Enhanced mobile broadband (eMBB)
- Ultra-reliable low-latency communication (URLLC)
- Massive machine-type communication (mMTC)
- Smart cities and intelligent transportation
- Industrial automation
- Advanced IoT deployments

4.3.5 Comparison 2G, 3G, 4G and 5G

Feature	2G	3G	4G (LTE)	5G
Primary Standard	GSM	UMTS	LTE	5G NR
Switching	Circuit Switched	Hybrid	All-IP (Packet)	All-IP (Slicing)
Multiple Access	TDMA / CDMA	WCDMA	OFDMA	OFDMA
Max Data Speed	~384 kbps	~2 Mbps	~1 Gbps	~20 Gbps
Latency	>500 ms	~100 ms	~20 ms	< 1 ms
Primary Service	Voice & SMS	Mobile Internet	HD Broadband	IoT & Smart Apps

Review Questions

Key Telecommunication Devices

1. Draw the block diagram of a telephone set and explain the function of each block.
2. State functions of following blocks of a telephone set: (a) Line Interface (Tip and Ring), (b) Ringer Circuit, (c) Hook Switch, (d) Dialing Circuit, (e) Microphone, (f) Receiver, (g) The Equalizer Circuit, (h) Hybrid Network
3. State and explain functions of a telephone set.

4. Draw the internal block diagram of a basic mobile phone (mobile station) and explain the function of each block.
5. List the blocks of transmitter section of a basic mobile phone and explain their functions.
6. List the blocks of receiver section of a basic mobile phone and explain their functions.
7. Describe the working principle of a mobile station.
8. What is modem? State and explain its functions.
9. Draw the block diagram of a modem and explain the function of each block.
10. List the blocks of transmitter section of a modem and explain their functions.
11. List the blocks of receiver section of a modem and explain their functions.
12. Describe the working principle of a modem.
13. State applications of a modem.

Telecommunication Standards

14. Define 'Telecommunication Standards' and explain their necessity in the modern global communication system.
15. Describe the primary functions of the ITU.
16. List the three main sectors of the ITU and state the function of each sector.
17. Explain the role of the IEEE in the communication systems.
18. List any three IEEE 802 series standards and state what each defines.
19. Explain the role of 3GPP in communication systems.
20. Compare ITU, IEEE and 3GPP based on their role, scope, standards developed, and areas of application.

Basics of 2G, 3G, 4G, and 5G

21. State three important features each of the following mobile communication technologies: (a) 2G, (b) 3G, (c) 4G, (d) 5G
22. State two limitations each of the following mobile technologies: (a) 2G, (b) 3G, (c) 4G

23. List the applications of the following mobile communication technologies: (a) 2G, (b) 3G, (c) 4G, (d) 5G
24. State any three advantages of: (a) 5G over 4G, (b) 4G over 3G, (c) 3G over 2G
25. Compare 2G, 3G, 4G, and 5G technologies based on the following parameters in tabular form: (a) Switching Technique, (b) Multiple Access Method, (c) Peak Data Speed, (d) Latency, (e) Core Technology/Standard, (f) Primary Services/Applications